Empirical Proof Record: VALIDATED

2026-05-23T23:09:42.131620+00:00

Empirical Proof Record — VALIDATED

Proof ID: `29ce31f6f8b1d37c6b8700f33c567034…`
Created: 2026-05-23T23:09:42.131620+00:00

1. Claim

> The posterior 94% HDI width on μ_Φ contracts at the theoretical $\sqrt{}$N rate as the number of observations grows. Specifically, HDI_width(N=200) / HDI_width(N=20) $\leq$ 0.40 (theoretical contraction factor $\sqrt{}$(20/200) = 0.316; the 0.40 ceiling allows finite-sample slack).

- **Operationalisation:** Fit a PyMC NUTS posterior to subsamples of the Φ trajectory at sizes 20, 50, 100, 200. Posterior model: μ ~ Normal(prior), σ ~ HalfNormal, y_i ~ Normal(μ, σ). Compute the 94% HDI width on μ at each size; the contraction metric is HDI_width(N=200) / HDI_width(N=20).
- **Threshold:** `hdi_contraction_ratio <= 0.4 ratio`
- **H0:** hdi_contraction_ratio > 0.40 (posterior fails to tighten at the predicted rate — data may be non-Gaussian, non-stationary, or sampler miscalibrated)
- **H1:** hdi_contraction_ratio <= 0.40 (posterior tightens approximately at the theoretical $\sqrt{}$N rate, confirming Bayesian inference behaves correctly on the Φ trajectory)

2. Verdict

VALIDATED — observed `0.27735050659795846`
HDI width N=20 → N=200: 0.0654 → 0.0181; contraction ratio 0.2774 (theoretical 0.3162; ceiling 0.4000). Empirical Φ mean=0.6140 sd=0.0680 (n=200).

3. Evidence

Pillar	Statistic	Value	95% CI	Library
pymc_nuts_posterior	hdi_contraction_ratio	0.27735050659795846	—	pymc 6.0.0

4. Pre-registration

- Registered at: 2026-05-23T23:09:41.906890+00:00
- Config hash: a47e4fd070e7935308f57886af093806…
- Data hash: 9ef317fe291445319fcba522d29836dc…

5. Signature

64a1ff7149b07aeffa264c584264066f61a7a961d9e78ad6746c1464f6e6d996